Q2. Gradient Descent and the Impact of Learning Rate on Convergence

Gradient Descent is an optimization algorithm used in machine learning to minimize the loss function by iteratively updating model parameters in the direction that reduces the error. It works by calculating the gradient, which indicates the direction and rate at which the loss function increases, and then updating the parameters in the opposite direction of the gradient.

The parameter update rule is:

$$\theta_{new} = \theta_{old} - \eta \times \nabla J(\theta)$$

Here, θ represents the model parameters, η (eta) is the learning rate, and ∇J(θ) is the gradient of the loss function. The learning rate determines the size of each parameter update and plays a crucial role in convergence.

If the learning rate is too small, the algorithm takes tiny steps toward the minimum, resulting in slow convergence and requiring many iterations. If the learning rate is appropriately chosen, the loss decreases efficiently and the parameters approach a minimum in fewer iterations. However, if the learning rate is too large, the algorithm may overshoot the minimum, oscillate around it, or diverge, causing the loss to increase instead of decrease.

Therefore, selecting an appropriate learning rate is essential for efficient and stable training. Techniques such as learning-rate scheduling, feature scaling, and adaptive optimizers like Adam can help improve convergence.